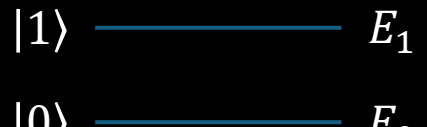
PHYS 209: Aspects of Modern Physics

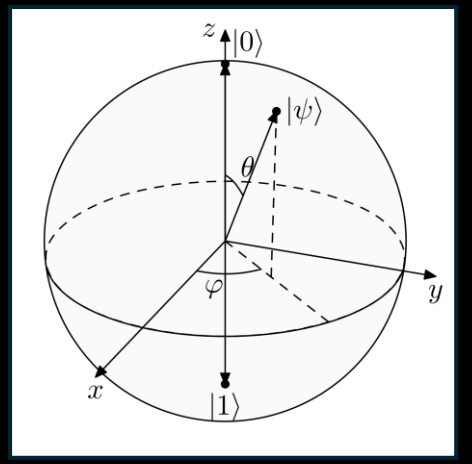
Module 4: Qubits

Lecture 3: Quantum Entanglement

Instructor: Muhammad Hani Zaheer

Review





- Qubits can be in superposition states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

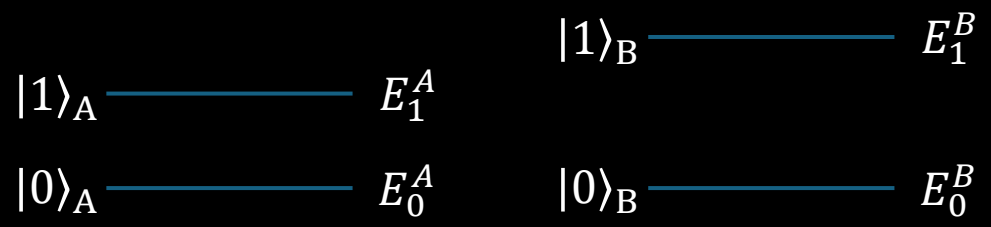
- Bloch Sphere Representation

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi}|1\rangle$$

- Manipulating Qubits = 2×2 matrices called operators
  - Examples: Identity + Pauli Matrices

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Two Qubits



- Consider two qubits

$$|\psi_A\rangle = \alpha_A|0\rangle_A + \beta_A|1\rangle_A, \qquad |\psi_B\rangle = \alpha_B|0\rangle_B + \beta_B|1\rangle_B$$

- The combined two-qubit state is obtained by performing a "tensor product"

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle = [\alpha_A|0\rangle_A + \beta_A|1\rangle_A] \otimes [\alpha_B|0\rangle_B + \beta_B|1\rangle_B]$$

$$|\psi\rangle = \alpha_A\alpha_B|0\rangle_A \otimes |0\rangle_B + \alpha_A\beta_B|0\rangle_A \otimes |1\rangle_B + \beta_A\alpha_B|1\rangle_A \otimes |0\rangle_B + \beta_A\beta_B|1\rangle_A \otimes |1\rangle_B$$

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

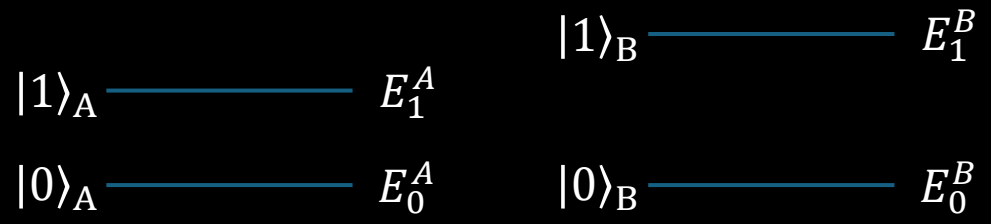
  - $|ij\rangle = |i\rangle \otimes |j\rangle$ denotes the state where qubit A is in $|i\rangle$ and qubit B is in $|j\rangle$.
- Here:

$$\alpha_{00} = \alpha_A\alpha_B, \qquad \alpha_{01} = \alpha_A\beta_B, \qquad \alpha_{10} = \beta_A\alpha_B, \qquad \alpha_{11} = \beta_A\beta_B$$

- Also note that:

$$\alpha_{00}\alpha_{11} = \alpha_{01}\alpha_{10}$$

Two Qubits States

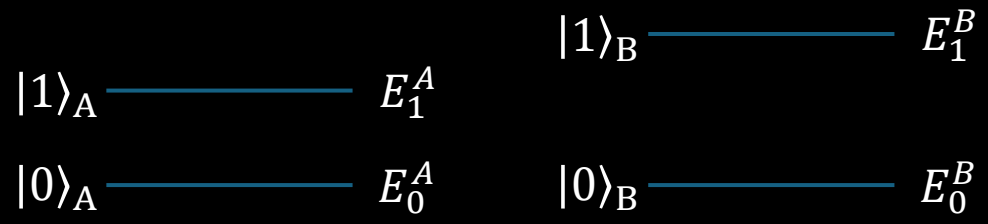


$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

- Probabilities:
  - $|\alpha_{ij}|^2$ gives probability of $|ij\rangle = |i\rangle \otimes |j\rangle$, meaning the probability that qubit A is measured to be in $|i\rangle$ and qubit B is in $|j\rangle$.
- Normalization:

$$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$$

Product States



- The following is an example of a product state:

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

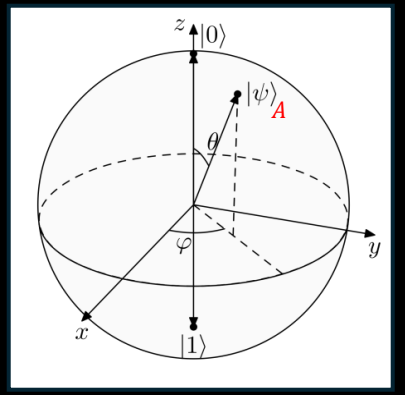
- In $|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$ representation, product states satisfy

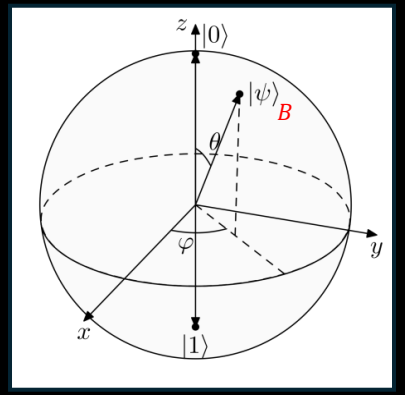
$$\alpha_{00}\alpha_{11} = \alpha_{01}\alpha_{10}$$

- Example:

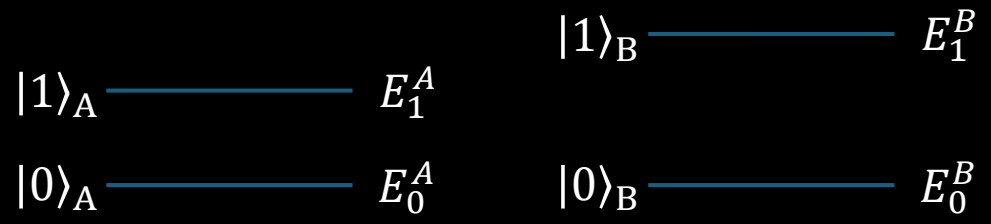
$$|\psi\rangle = |0\rangle \otimes |+\rangle = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{|00\rangle + |01\rangle}{\sqrt{2}}$$

- In a product state, each qubit has its own well-defined state — they are independent.
  - Measuring one tells you nothing about the other.
- We can use the Bloch sphere representation for the individual qubits.





Entangled States



- There are two-qubit states that cannot be written in product form
$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$
  - These are called entangled states.
- In the basis $|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$, entangled states are characterized by
$$\alpha_{00}\alpha_{11} \neq \alpha_{01}\alpha_{10}$$
- Example:
$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$
  - We cannot find a decomposition of this state into $|\psi_A\rangle$ and $|\psi_B\rangle$.
  - We can prove this by contradiction.
  - Or we can prove this by noting $\alpha_{00} = \alpha_{11} = 1$ and $\alpha_{01} = \alpha_{10} = 0$.

Proof: Entangled States ≠ Product States

- Consider two-qubit state

$$|\psi_A\rangle = \alpha_A|0\rangle_A + \beta_A|1\rangle_A, \qquad |\psi_B\rangle = \alpha_B|0\rangle_B + \beta_B|1\rangle_B$$

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

$$\alpha_{00} = \alpha_A\alpha_B, \qquad \alpha_{01} = \alpha_A\beta_B, \qquad \alpha_{10} = \beta_A\alpha_B, \qquad \alpha_{11} = \beta_A\beta_B$$

- Consider the entangled state

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- We can attempt to find the individual state representation by matching coefficients:

$$\alpha_{00} = \frac{1}{\sqrt{2}}, \qquad \alpha_{01} = 0, \qquad \alpha_{10} = 0, \qquad \alpha_{11} = \frac{1}{\sqrt{2}}$$

$$\alpha_A\alpha_B = \frac{1}{\sqrt{2}}, \qquad \alpha_A\beta_B = 0, \qquad \beta_A\alpha_B = 0, \qquad \beta_A\beta_B = \frac{1}{\sqrt{2}}$$

Proof: Entangled States ≠ Product States

$$\alpha_A\alpha_B = \frac{1}{\sqrt{2}}, \qquad \alpha_A\beta_B = 0, \qquad \beta_A\alpha_B = 0, \qquad \beta_A\beta_B = \frac{1}{\sqrt{2}}$$

- If $\alpha_A\beta_B = 0$, then
  - Either $\alpha_A = 0$
    - Which contradicts $\alpha_A\alpha_B = \frac{1}{\sqrt{2}}$
  - Or $\beta_B = 0$
    - Which contradicts $\beta_A\beta_B = \frac{1}{\sqrt{2}}$
- Thus, we necessarily have a contradiction when we attempt to match coefficients.
- Therefore, $|\phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ cannot be written as a product state.

Measurement in Entangled States

- Suppose Alice makes a measurement along z

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \xrightarrow{\text{Alice measures } |0\rangle} |00\rangle \rightarrow \text{Bob has } |0\rangle$$

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \xrightarrow{\text{Alice measures } |1\rangle} |11\rangle \rightarrow \text{Bob has } |1\rangle$$

- Perfect correlation: Alice and Bob always get the same outcome
  - Alice's individual outcomes each have 50% probability.
  - But Bob's outcomes along the same axis are determined after Alice's measurement.
  - Bob's outcomes along different axis are completely random.
- This holds regardless of distance between Alice and Bob!

Measurement in Entangled States

- Consider eigenstates of $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- Inverting the basis:

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \qquad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}$$

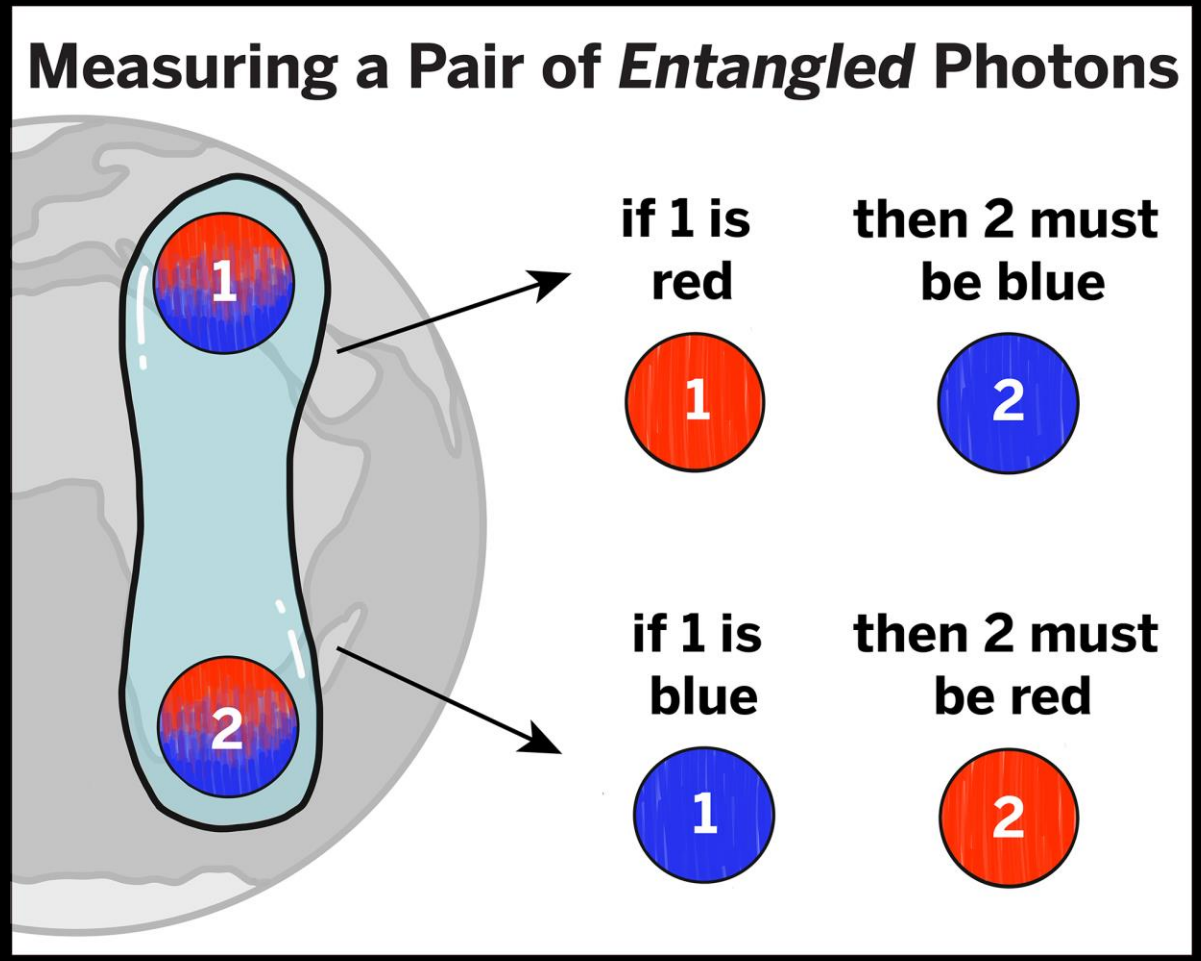
- The entangled state can be written in X basis

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\left[\frac{|+\rangle + |-\rangle}{\sqrt{2}} \otimes \frac{|+\rangle + |-\rangle}{\sqrt{2}} + \frac{|+\rangle - |-\rangle}{\sqrt{2}} \otimes \frac{|+\rangle - |-\rangle}{\sqrt{2}}\right]$$

$$= \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$$

- Perfect correlation in x as well: Alice and Bob get same outcome
- In fact, perfect correlation holds in any basis!

Measurement in Entangled States



https://quantumatlas.umd.edu/entry/entanglement/

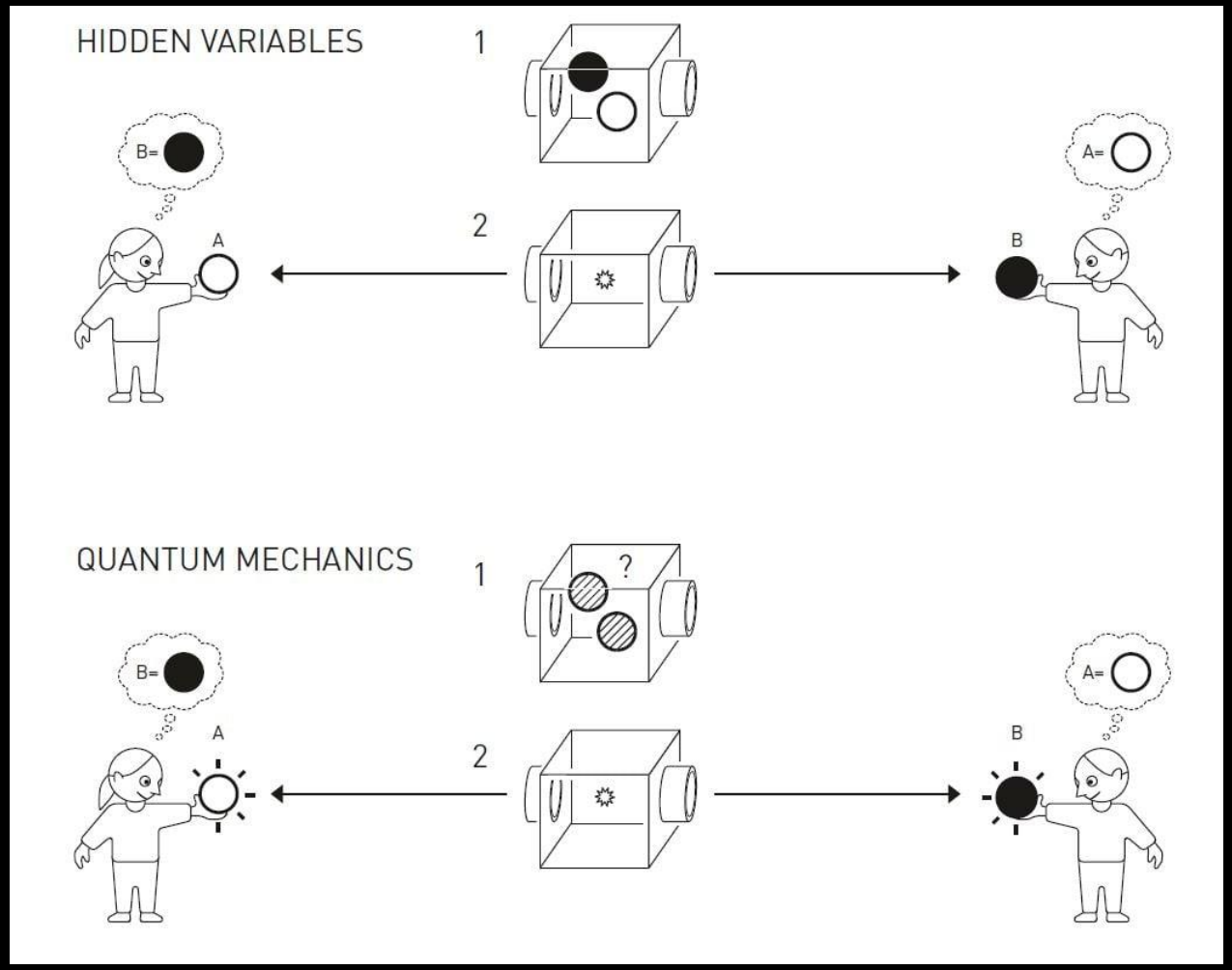
Einstein-Podolsky-Rosen (EPR) Paradox

- Alice cannot affect Bob (locality).
  - No “spooky action at a distance”.
- Therefore, Bob’s state must be predetermined (realism).
- This must be true along any axis since Bob does not know which axis Alice will perform her measurement along.
- If X and Z eigenvalues are simultaneously predetermined, there must be hidden variables that encode this information.
- But X and Z eigenvalues cannot be simultaneously known in Quantum Mechanics since

$$[\hat{Z},\hat{X}] = \hat{Z}\hat{X} - \hat{X}\hat{Z} = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} - \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}0 & 2\\ -2 & 0\end{bmatrix} = 2i\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix} = 2i\hat{Y}$$

- This is the EPR paradox: locality + QM predictions force hidden variables, but QM itself forbids them. Either local realism or QM must be wrong!

Hidden Variables vs Quantum Mechanics



https://www.nobelprize.org/prizes/physics/2022/popular-information/

EPR Paradox Summary

- Locality + QM Predictions → Hidden Variables
- But $[\hat{X}, \hat{Z}] \neq 0$ means hidden variables contradict QM
- Question: Is there any experiment where QM and Hidden variable theories give different predictions?

Correlation Function

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- Alice and Bob each receive one qubit
- Each independently chooses a measurement setting:
  - Alice choose direction of measurement $\hat{a}$ or $\hat{a}'$
  - Bob chooses direction of measurement $\hat{b}$ or $\hat{b}'$
- Each measurement outcome is ± 1
- Let us define:
  - $A(\hat{a}) = \vec{\sigma} \cdot \hat{a}$: Measurement outcome of Alice's measurement along $\hat{a}$
  - $B(\hat{b}) = \vec{\sigma} \cdot \hat{b}$: Measurement outcome of Bob's measurement along $\hat{b}$
- Example: $A(\hat{z}) = \hat{Z}$, $B(\hat{x}) = \hat{X}$

Correlation Function

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- We define the correlation function:

$$E(\hat{a}, \hat{b}) = \langle A(\hat{a}) \cdot B(\hat{b}) \rangle$$

  - Local realism: Each measurement outcome is predetermined
  - QM: Each measurement outcome is random
- It turns out that in Bell state $|\phi^+\rangle$:
  - Local realism predicts $|E(\hat{a}, \hat{b})| \leq 1$
  - Quantum mechanics predicts $|E(\hat{a}, \hat{b})| \leq 1$
- Identical predictions for a single correlation.
- What about multiple correlations?

Multiple Correlations

- Alice chooses between two measurement directions: $\hat{a}$ or $\hat{a}'$
- Bob chooses between two measurement directions: $\hat{b}$ or $\hat{b}'$
- This gives four correlations:

$$E(\hat{a},\hat{b}), \qquad E(\hat{a},\hat{b}'), \qquad E(\hat{a}',\hat{b}), \qquad E(\hat{a}',\hat{b}')$$

- Let us define the combination

$$S = E(\hat{a},\hat{b}) - E(\hat{a},\hat{b}') + E(\hat{a}',\hat{b}) + E(\hat{a}',\hat{b}')$$

$$S = A(\hat{a})B(\hat{b}) - A(\hat{a})B(\hat{b}') + A(\hat{a}')B(\hat{b}) + A(\hat{a}')B(\hat{b}')$$

$$S = A(\hat{a})\left[B(\hat{b}) - B(\hat{b}')\right] + A(\hat{a}')\left[B(\hat{b}) + B(\hat{b}')\right]$$

CHSH Inequality (Local Realism)

$$S = A(\hat{a})\left[B(\hat{b}) - B(\hat{b}')\right] + A(\hat{a}')\left[B(\hat{b}) + B(\hat{b}')\right]$$

- Remember: $B(\hat{b}) = \pm 1$ and $B(\hat{b}') = \pm 1$

- Case 1: $B(\hat{b}) = B(\hat{b}') = \pm 1$

$$S = A(\hat{a}') \cdot (\pm 2) \rightarrow |S| = 2$$

- Case 2: $B(\hat{b}) = -B(\hat{b}') = \pm 1$

$$S = A(\hat{a}) \cdot (\pm 2) \rightarrow |S| = 2$$

- Taking average over many trials:

$$|\langle S \rangle| \leq \langle |S| \rangle = 2$$

Quantum Mechanics Prediction

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- Correlation Function:

$$E(\hat{a}, \hat{b}) = \langle A(\hat{a})B(\hat{b})\rangle = \langle \phi^+ | (\vec{\sigma}\cdot\hat{a}) \otimes (\vec{\sigma}\cdot\hat{b}) | \phi^+\rangle = \cos\theta_{ab}$$

- Choose angles to maximize $|S|$:

$$\theta_a = 0^\circ, \qquad \theta_b = 45^\circ, \qquad \theta_{a'} = 90^\circ, \qquad \theta_{b'} = 135^\circ$$

- Compute correlations:

$$E(\hat{a}, \hat{b}) = \cos 45^\circ = \frac{1}{\sqrt{2}}, \qquad E(\hat{a}, \hat{b}') = \cos 135^\circ = -\frac{1}{\sqrt{2}}$$

$$E(\hat{a}', \hat{b}) = \cos 45^\circ = \frac{1}{\sqrt{2}}, \qquad E(\hat{a}', \hat{b}') = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Quantum Mechanics Prediction

$$E(\hat{a},\hat{b}) = \frac{1}{\sqrt{2}}, \qquad E(\hat{a},\hat{b}') = -\frac{1}{\sqrt{2}}$$
$$E(\hat{a}',\hat{b}) = \frac{1}{\sqrt{2}}, \qquad E(\hat{a}',\hat{b}') = \frac{1}{\sqrt{2}}$$

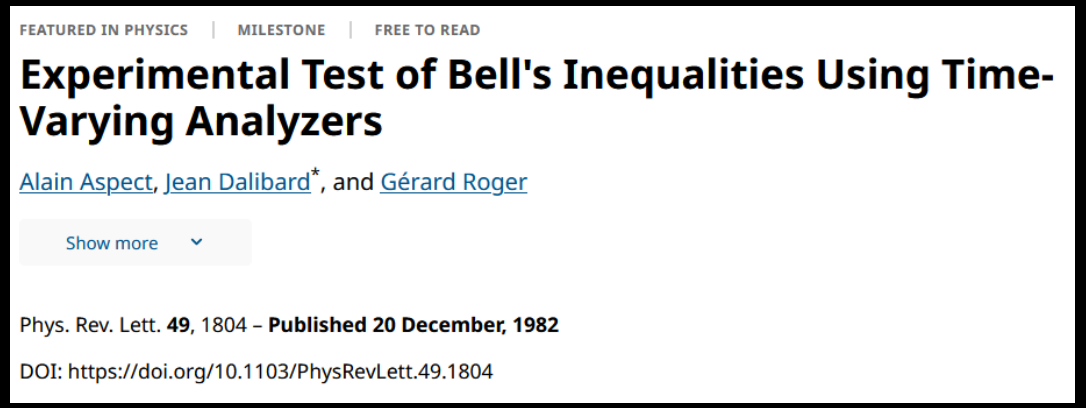
- Compute S:

$$S = E(\hat{a},\hat{b}) - E(\hat{a},\hat{b}') + E(\hat{a}',\hat{b}) + E(\hat{a}',\hat{b}')$$
$$S = \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$
$$|S|_{QM} = 2\sqrt{2} \approx 2.83 > 2$$

- Quantum Mechanics violates CHSH inequality!
- Bell's Theorem: No hidden variable theory satisfying local realism can reproduce all predictions of Quantum Mechanics.

Local Realism vs Quantum Mechanics

- Local Realism: $|S| \leq 2$
- Quantum mechanics: $|S| = 2\sqrt{2} \approx 2.83$
- Experiments agree with Quantum Mechanics!
  - Local Realism ruled out!



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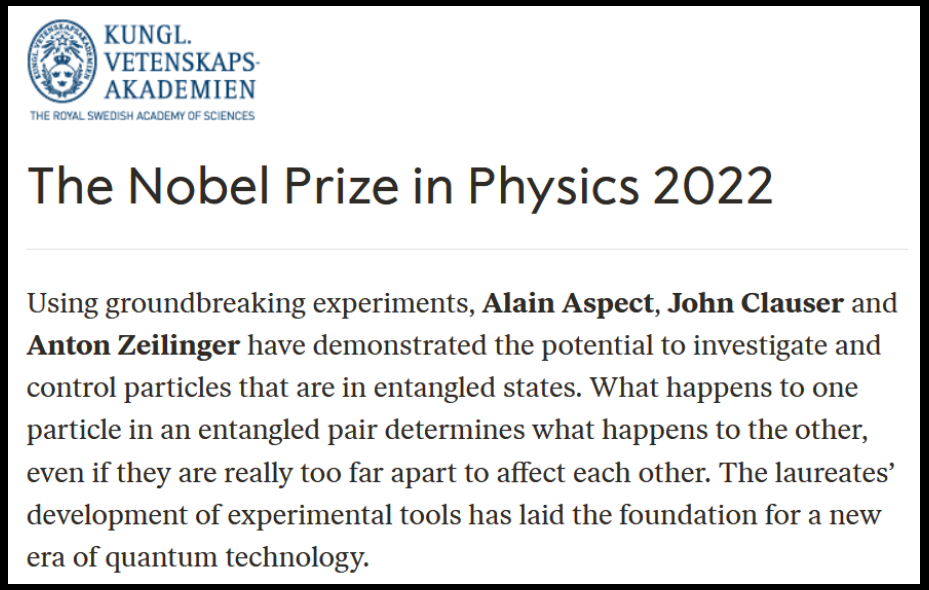
Experimental Test of Bell's Inequalities Using Time-Varying Analyzers

Alain Aspect, Jean Dalibard*, and Gérard Roger

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Phys. Rev. Lett. **49**, 1804 – **Published 20 December, 1982**

DOI: https://doi.org/10.1103/PhysRevLett.49.1804



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The Nobel Prize in Physics 2022

Using groundbreaking experiments, **Alain Aspect**, **John Clauser** and **Anton Zeilinger** have demonstrated the potential to investigate and control particles that are in entangled states. What happens to one particle in an entangled pair determines what happens to the other, even if they are really too far apart to affect each other. The laureates' development of experimental tools has laid the foundation for a new era of quantum technology.

Conclusion

- No realism: Measurement outcomes are intrinsically random – no hidden variables
- Weak locality: Entanglement cannot be used to send information faster than light
  - Bob's outcomes are always 50% regardless of Alice's measurement choice
  - Correlations only appear when Alice & Bob meet and compare results
  - Classical comparison cannot exceed the speed of light

$$\text{Entanglement} \neq \text{Communication}$$

- However, hidden variable theories are technically not ruled out as long as they abandon strong locality!
- Bottom line:

Nature is not locally real.